Physicists describe these two properties of physical laws—that they do not depend on when or where you use them—as symmetries of nature. By this usage physicists mean that nature treats every moment in time and every location in space identically—symmetrically—by ensuring that the same fundamental laws are in operation. Much in the same manner that they affect art and music, such symmetries are deeply satisfying; they highlight an order and a coherence in the workings of nature. The elegance of rich, complex, and diverse phenomena emerging from a simple set of universal laws is at least part of what physicists mean when they invoke the term “beautiful.”

In our discussions of the special and general theories of relativity, we came upon yet other symmetries of nature. Recall that the principle of relativity, which lies at the heart of special relativity, tells us that all physical laws must be the same regardless of the constant-velocity relative motion that individual observers might experience. This is a symmetry because it means that nature treats all such observers identically—symmetrically. Each such observer is justified in considering himself or herself to be at rest. Again, it’s not that observers in relative motion will make identical observations; as we have seen earlier, there are all sorts of stunning differences in their observations. Instead, like the disparate experiences of the pogo-stick enthusiast on the earth and on the moon, the differences in observations reflect environmental details—the observers are in relative motion—even though their observations are governed by identical laws.

Through the equivalence principle of general relativity, Einstein significantly extended this symmetry by showing that the laws of physics are actually identical for all observers, even if they are undergoing complicated accelerated motion. Recall that Einstein accomplished this by realizing that an accelerated observer is also perfectly justified in declaring himself or herself to be at rest, and in claiming that the force he or she feels is due to a gravitational field. Once gravity is included in the framework, all possible observational vantage points are on a completely equal footing. Beyond the intrinsic aesthetic appeal of this egalitarian treatment of all motion, we have seen that these symmetry principles played a pivotal role in the stunning conclusions regarding gravity that Einstein found.